MACHINE LEARNING · REGRESSION

Multiple Linear Regression: From Concept to Code

Bridging the statistics deck with real scripts, data, and results

THE PROBLEM

One Predictor Isn't Enough

TV alone only explains ~61% of sales variance (previous deck)
We also have radio and newspaper budgets
Are they independently useful, or just noise once TV is in the model?
Multiple regression lets us ask that question properly

THE MODEL

Sales = $\beta 0 + \beta 1 \cdot TV + \beta 2 \cdot Radio + \beta 3 \cdot Newspaper + e$

Same least-squares idea as before, now with p predictors instead of 1
Coefficients are estimated jointly
Each β_j answers: “what's the effect of this channel, holding the others fixed?”

Is There a Relationship At All?

Tempting shortcut: run a separate t-test per predictor
Problem: with many predictors, some look significant purely by chance (multiple-testing inflation)
Fix: the F-statistic tests all predictors jointly — “is at least one related to sales?” — before inspecting individual ones

THE CODE

Code: Load & Split

Advertising.csv → X = [TV, radio, newspaper], y = sales
train_test_split(X, y, test_size=0.2, random_state=42)
Fit only on the training 80%, evaluate on the held-out 20%

Code: Fit the Model

model = LinearRegression().fit(X_train, y_train)
Intercept ≈ 2.98
Radio's coefficient is ~4× TV's, and ~70× newspaper's — per dollar, radio moves sales the most

Term	Coefficient
Intercept	2.979
TV	0.0447
Radio	0.1892
Newspaper	0.0028

Evaluating the Fit

R^2 always increases (or stays flat) as you add predictors — even useless ones
Don't trust R^2 alone — compare against test-set error too
Our own R^2 as predictors are added, computed from this data:

Predictors	R^2
TV only	0.612
TV + Radio	0.897
TV + Radio + Newspaper	0.897

Code: Evaluation on the Test Set

Predict on the untouched 20% test split

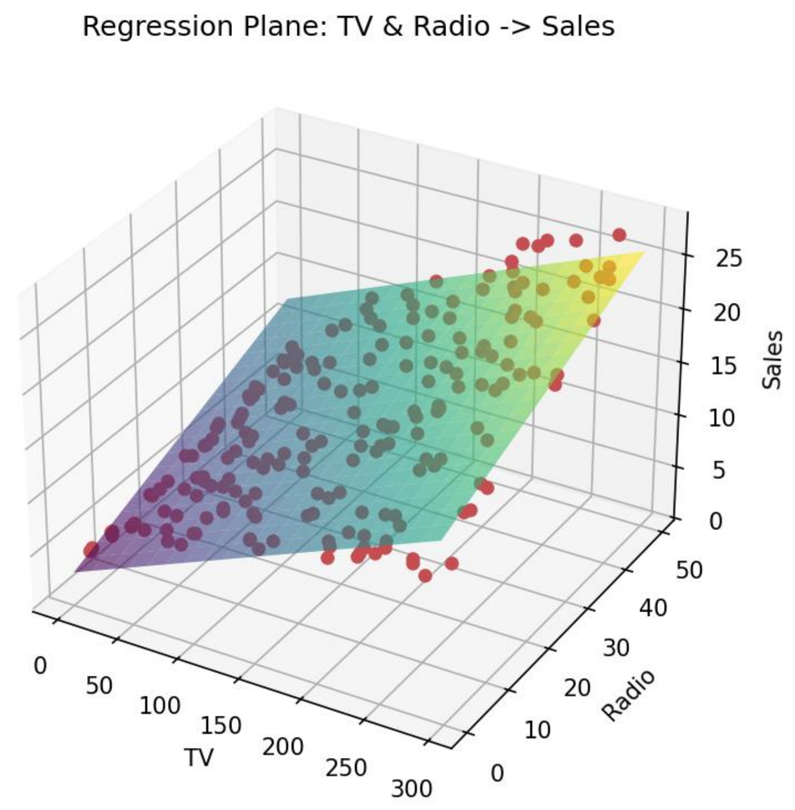
Test $R^2 \approx 0.899$, Test MSE ≈ 3.17

Newspaper's near-zero R^2 contribution above matches the test set — it adds almost nothing once TV & radio are in the model

Visualizing in 3D

Two predictors (TV, Radio) let us actually see the fitted plane against Sales

Points off the plane are residual error



A REAL LIMITATION

The Synergy Question

The theory deck flags a real limitation: an additive model assumes TV and radio act independently

Observed pattern: combined spending outperforms what the additive model predicts — a synergy / interaction effect

Theory's fix: $\text{Sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{Radio} + \beta_3 \cdot (\text{TV} \times \text{Radio})$

Gap: our script does not yet fit this interaction term

A REAL LIMITATION

Confidence vs. Prediction Intervals

Theory deck distinguishes: CI = uncertainty in the average response; PI = uncertainty in one new observation (PI is wider)

Gap: our script currently reports a single point prediction only — no CI/PI

Where the Script Stands vs. the Full Theory

Concept	In the theory deck	In our script
F-test for overall significance	Yes	Not yet
Individual coefficients & interpretation	Yes	Yes
R^2 / test-set evaluation	Yes	Yes
Interaction (synergy) term	Yes	Not yet
Confidence / prediction intervals	Yes	Not yet
Variable selection (forward/backward /mixed)	Yes	Not yet

Key Takeaways

More predictors need joint estimation, not p separate simple regressions
R^2 alone can mislead — the F-test and held-out test error tell the real story
Our script covers fitting + evaluation solidly; interaction terms, CI/PI, and formal variable selection are the clear next steps
Full code: Multiple_linear_regression2.ipynb